AMIGA

Learn which network to choose.

Automated Multi-objective Individual GRN Assessment

From alternative consensus networks to a ranked recommendation — without reference labels at prediction time.

UPSTREAM / GENERATE ALTERNATIVES | **AMIGA / DESCRIBE, LEARN AND RANK**

1 Learn from benchmark networks

Known reference links reveal which candidates deserve higher priority.

Benchmark data

Expression + base networks

Gene expression | Base GRNs

Reference network
Known links

Explore candidate networks

Multi-objective consensus optimization

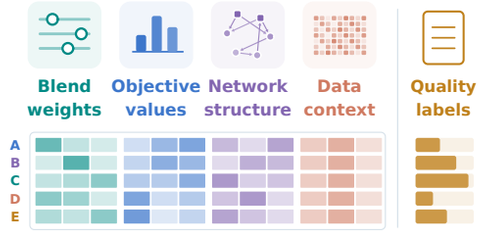


One point = one weighted consensus.
The Pareto front contains trade-offs.

Compare with reference links
Measured quality label (AUPR)

Describe each candidate

One table row per candidate network



Network descriptors vary by candidate; expression context is shared by a front.

Learn to rank within each front

Features predict an ordering, not a cutoff



Prefer candidates with higher quality labels

Grouped validation
Keep fronts intact



Save the ranker + feature schema
Refit on all labelled fronts after CV

Quality labels supervise learning; they are not prediction inputs.

Reuse the learned decision rule
Trained model + saved predictor columns

2 Recommend a network for new data

The learned decision rule scores new candidates without a reference network.

New study data

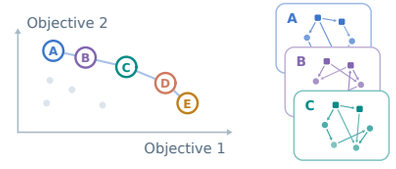
Expression + base networks

Gene expression | Base GRNs

No reference network is needed here
Only experimental inputs

Explore candidate networks

Multi-objective consensus optimization

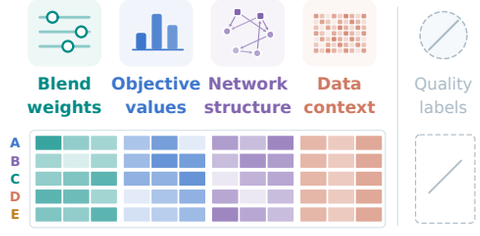


One point = one weighted consensus.
The Pareto front contains trade-offs.

Several alternatives.
No reference-based quality label.

Describe each candidate

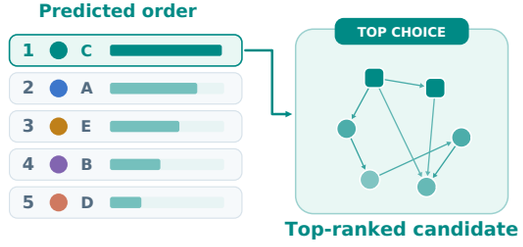
One table row per candidate network



Use the same predictor columns and order saved during training.

Score, rank and recommend

Apply the saved AMIGA model



Top-ranked candidate

Choose one network or a short list

Output: score + rank within each front

AMIGA prioritizes existing candidates; it does not generate a new network.

Schematics, not experimental results. Ranking scores are not measured network accuracy.

AMIGA core workflow